

Chapter 4: Fractions and percentages

12 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems.

CHAPTER

4

4.01 Fractions

4.02 Ordering fractions

4.03 Adding and subtracting fractions

4.04 Adding and subtracting mixed numerals

4.05 Fraction of a quantity

4.06 Multiplying fractions

4.07 Dividing fractions

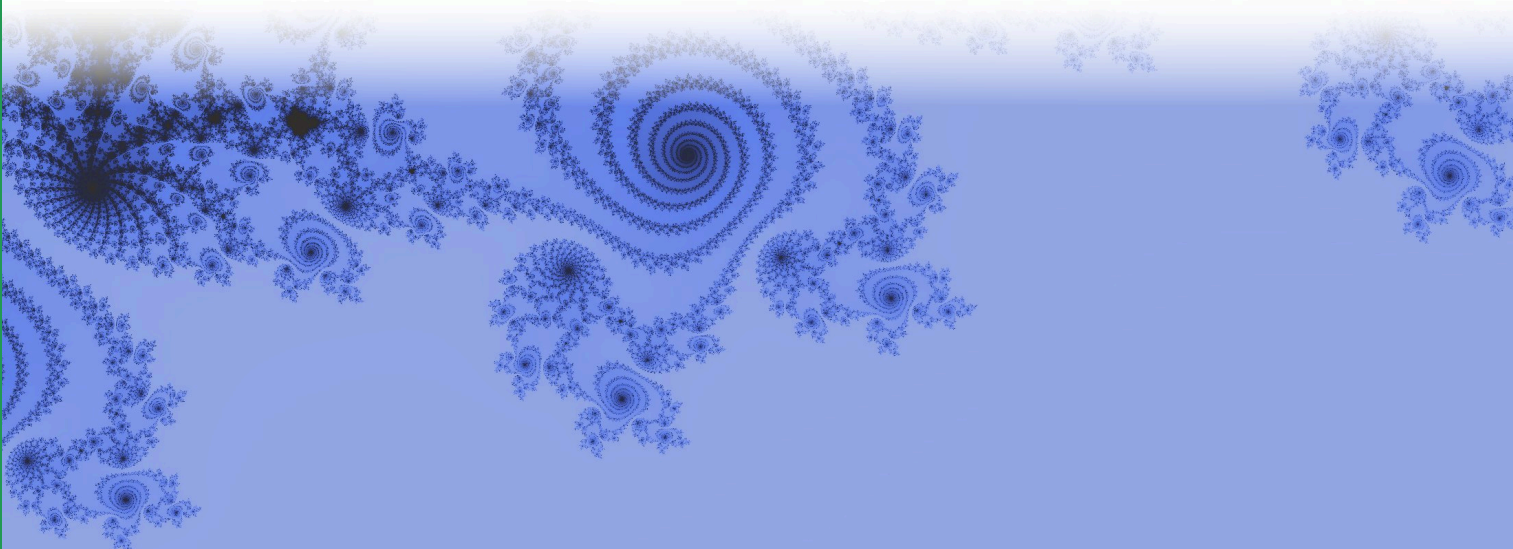
4.08 Fraction problems

4.09 Fractions on a calculator

4.10 Percentages, fractions and decimals

4.11 Percentage of a quantity

4.12 Expressing as a fraction or percentage



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

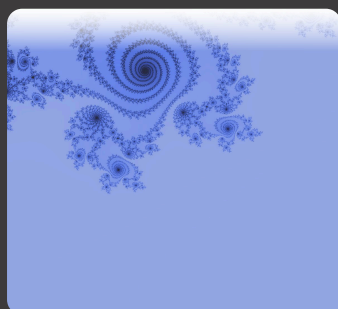
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
4.01	Fractions	3		☐
4.02	Ordering fractions	9		☐
4.03	Adding and subtracting fractions	15		☐
4.04	Adding and subtracting mixed numerals	21		☐
4.05	Fraction of a quantity	27		☐
4.06	Multiplying fractions	33		☐
4.07	Dividing fractions	39		☐
4.08	Fraction problems	45		☐
4.09	Fractions on a calculator	53		☐
4.10	Percentages, fractions and decimals	59		☐
4.11	Percentage of a quantity	65		☐
4.12	Expressing as a fraction or percentage	71		☐

In this chapter you will

- write equivalent fractions and simplify
- order fractions
- add, subtract, multiply and divide fractions and mixed numerals
- find a fraction or percentage of a quantity
- solve fraction problems
- convert between fractions, decimals and percentages



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a triple spiral on the edge of the top bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
4.01

Fractions

Date _____

Learning intentions

We are learning to:

- convert between mixed numerals and improper fractions
- write equivalent fractions and simplify fractions

Success criteria

I can:

- ☐ change a mixed numeral to an improper fraction and back
- ☐ multiply top and bottom by the same number to find an equivalent fraction
- ☐ divide top and bottom by the HCF to simplify

Key terms Write the definition as your teacher explains it.

Improper fraction

Mixed numeral

Equivalent

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Convert between mixed numerals and improper fractions.**a** Write $1\frac{3}{4}$ as an improper fraction.**b** Write $\frac{7}{4}$ as a mixed numeral.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write each mixed numeral as an improper fraction.

12 QUESTIONS

REMEMBER Mixed → improper: whole × bottom + top.

1 $1\frac{3}{4}$

2 $2\frac{4}{9}$

3 $5\frac{3}{5}$

4 $3\frac{1}{7}$

5 $4\frac{1}{2}$

6 $2\frac{5}{6}$

Questions 7–12: simplify each fraction.

7 $\frac{6}{8}$

8 $\frac{10}{15}$

9 $\frac{12}{16}$

10 $\frac{9}{27}$

11 $\frac{14}{21}$

12 $\frac{20}{50}$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: simplify each fraction.

1 $\frac{5}{10}$

2 $\frac{66}{72}$

3 $\frac{2}{6}$

4 $\frac{3}{9}$

5 $\frac{6}{12}$

6 $\frac{3}{6}$

7 $\frac{4}{48}$

8 $\frac{12}{40}$

9 $\frac{10}{16}$

10 $\frac{6}{18}$

11 $\frac{55}{60}$

12 $\frac{54}{60}$

13 $\frac{6}{9}$

14 $\frac{15}{25}$

15 $\frac{4}{40}$

16 $\frac{10}{12}$

17 $\frac{3}{12}$

18 $\frac{9}{15}$

19 $\frac{15}{20}$

20 $\frac{15}{24}$

21 $\frac{9}{24}$

22 $\frac{6}{8}$

23 $\frac{12}{18}$

24 $\frac{5}{40}$

ROUND 2

Questions 25–40: change mixed numerals to improper fractions, and improper fractions to mixed numerals.

25 $5\frac{2}{3}$

26 $2\frac{1}{10}$

27 $\frac{7}{2}$

28 $6\frac{1}{6}$

29 $6\frac{2}{5}$

30 $\frac{8}{3}$

31 $5\frac{7}{10}$

32 $\frac{53}{12}$

33 $4\frac{1}{4}$

34 $1\frac{2}{3}$

35 $1\frac{9}{10}$

36 $\frac{19}{3}$

37 $3\frac{3}{5}$

38 $\frac{17}{5}$

39 $\frac{11}{2}$

40 $\frac{3}{2}$

YOU DO

On your own: **Easy** Set B. Write each improper fraction as a mixed numeral.

6 QUESTIONS

1 $\frac{7}{4}$

4 $\frac{11}{2}$

2 $\frac{26}{3}$

5 $\frac{17}{5}$

3 $\frac{23}{7}$

6 $\frac{36}{24}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Complete each pair of equivalent fractions.

a Complete: $\frac{1}{3} = \frac{?}{15}$

Your teacher works this out. Copy every step.

b Complete: $\frac{3}{4} = \frac{12}{?}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Complete each pair of equivalent fractions.

6 QUESTIONS

REMEMBER Improper → mixed: divide top by bottom.

1 $\frac{1}{3} = \frac{?}{15}$

3 $\frac{4}{5} = \frac{?}{20}$

5 $\frac{1}{2} = \frac{?}{50}$

2 $\frac{3}{4} = \frac{12}{?}$

4 $\frac{5}{12} = \frac{15}{?}$

6 $\frac{3}{20} = \frac{?}{100}$

SKILL DRILL 2 Medium basics: keep practising

36 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: find the missing number in each pair of equivalent fractions.

1 $\frac{5}{8} = \frac{30}{?}$

2 $\frac{3}{4} = \frac{21}{?}$

3 $\frac{2}{5} = \frac{8}{?}$

4 $\frac{1}{6} = \frac{?}{54}$

5 $\frac{1}{2} = \frac{?}{8}$

6 $\frac{6}{10} = \frac{24}{?}$

7 $\frac{4}{10} = \frac{?}{30}$

8 $\frac{3}{4} = \frac{15}{?}$

9 $\frac{1}{2} = \frac{?}{4}$

10 $\frac{3}{4} = \frac{6}{?}$

11 $\frac{3}{6} = \frac{?}{48}$

12 $\frac{8}{12} = \frac{40}{?}$

13 $\frac{1}{2} = \frac{?}{10}$

14 $\frac{1}{3} = \frac{?}{9}$

15 $\frac{1}{3} = \frac{8}{?}$

16 $\frac{2}{3} = \frac{?}{9}$

17 $\frac{3}{6} = \frac{15}{?}$

18 $\frac{1}{8} = \frac{9}{?}$

19 $\frac{9}{10} = \frac{45}{?}$

20 $\frac{1}{2} = \frac{9}{?}$

ROUND 2 Questions 21–36: simplify fully.

21 $\frac{45}{90}$

22 $\frac{120}{180}$

23 $\frac{120}{135}$

24 $\frac{18}{27}$

25 $\frac{18}{60}$

26 $\frac{12}{180}$

27 $\frac{48}{80}$

28 $\frac{16}{72}$

29 $\frac{16}{64}$

30 $\frac{6}{54}$

31 $\frac{18}{54}$

32 $\frac{8}{48}$

33 $\frac{81}{99}$

34 $\frac{144}{180}$

35 $\frac{12}{24}$

36 $\frac{30}{48}$

Level 3 · Challenge

Challenging · 6 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Simplify each fraction.

a Simplify $\frac{25}{30}$.

b Simplify $\frac{36}{24}$ and write as a mixed numeral.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Simplify each fraction.

6 QUESTIONS

REMEMBER Equivalent: $\times$ or $\div$ top and bottom by the same number.

1 $\frac{25}{30}$

3 $\frac{15}{20}$

5 $\frac{24}{48}$

2 $\frac{30}{40}$

4 $\frac{20}{15}$

6 $\frac{24}{42}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write $3\frac{2}{5}$ as an improper fraction.

Level 2

Complete: $\frac{2}{7} = \frac{?}{35}$

Level 3

Simplify $\frac{45}{60}$.

Extension · Stretch yourself**Worked example**

Find three fractions equivalent to $\frac{2}{3}$ with denominators between 10 and 20.

1. Multiply top and bottom by the same number
2. $\times 4$, $\times 5$, $\times 6$

Answer: $\frac{8}{12}$, $\frac{10}{15}$, $\frac{12}{18}$

YOU DO

On your own: read the worked example, then try these.

- E1 Which fraction is not equivalent to $\frac{3}{5}$: $\frac{9}{15}$, $\frac{12}{20}$, $\frac{15}{30}$, $\frac{30}{50}$?

- E2 A fraction simplifies to $\frac{2}{7}$. Its numerator and denominator add to 45. Find the fraction.

Lesson summary

Missed the lesson? Start here.

Remember

1. Mixed $\rightarrow$ improper: whole $\times$ bottom + top.
2. Improper $\rightarrow$ mixed: divide top by bottom.
3. Equivalent: $\times$ or $\div$ top and bottom by the same number.
4. Simplify: $\div$ top and bottom by the HCF.

Level 1

$$1\frac{3}{4}$$

1. $1 \times 4 + 3 = 7$

Answer: $\frac{7}{4}$

Level 2

$$\frac{26}{3}$$

1. $26 \div 3 = 8 \text{ r } 2$

Answer: $8\frac{2}{3}$

Level 3

$$\text{Simplify } \frac{24}{42}$$

1. HCF is 6

Answer: $\frac{4}{7}$

Extension

$$\text{Equivalent to } \frac{2}{3}$$

1. $\times 4$, $\times 5$, $\times 6$

Answer: $\frac{8}{12}$, $\frac{10}{15}$, $\frac{12}{18}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.02

Ordering fractions

Date _____

Learning intentions

We are learning to:

- compare and order fractions

Success criteria

I can:

- ☐ compare fractions with the same denominator
- ☐ change fractions to a common denominator
- ☐ order fractions from smallest to largest

Key terms Write the definition as your teacher explains it.**Common denominator****Ascending order**

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Which fraction is larger?**a** Which is larger: $\frac{3}{8}$ or $\frac{5}{8}$?

Your teacher works this out. Copy every step.

b Which is larger: $\frac{2}{3}$ or $\frac{3}{5}$?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Which fraction is larger?

12 QUESTIONS

REMEMBER Same denominator: compare the numerators.

① $\frac{3}{7}$ or $\frac{5}{7}$

② $\frac{1}{4}$ or $\frac{1}{3}$

③ $\frac{9}{10}$ or $\frac{7}{10}$

④ $\frac{1}{2}$ or $\frac{2}{5}$

⑤ $\frac{2}{3}$ or $\frac{5}{6}$

⑥ $\frac{3}{4}$ or $\frac{5}{6}$

Questions 7–12: which is larger?

⑦ $\frac{2}{5}$ or $\frac{3}{5}$

⑧ $\frac{1}{2}$ or $\frac{3}{8}$

⑨ $\frac{2}{3}$ or $\frac{3}{4}$

⑩ $\frac{5}{6}$ or $\frac{7}{9}$

⑪ $\frac{3}{10}$ or $\frac{1}{4}$

⑫ $\frac{4}{7}$ or $\frac{1}{2}$

SKILL DRILL 1 Easy basics: practise until it is automatic

32 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 32

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: which fraction is larger? The denominators (or numerators) are the same.

① $\frac{2}{4}$ or $\frac{3}{4}$

② $\frac{7}{10}$ or $\frac{5}{10}$

③ $\frac{1}{5}$ or $\frac{1}{8}$

④ $\frac{6}{10}$ or $\frac{5}{10}$

⑤ $\frac{5}{6}$ or $\frac{5}{12}$

⑥ $\frac{6}{10}$ or $\frac{9}{10}$

⑦ $\frac{11}{12}$ or $\frac{2}{12}$

⑧ $\frac{4}{11}$ or $\frac{4}{9}$

⑨ $\frac{4}{12}$ or $\frac{4}{10}$

⑩ $\frac{1}{10}$ or $\frac{1}{2}$

⑪ $\frac{2}{5}$ or $\frac{4}{5}$

⑫ $\frac{3}{11}$ or $\frac{3}{7}$

⑬ $\frac{9}{10}$ or $\frac{8}{10}$

⑭ $\frac{2}{4}$ or $\frac{2}{11}$

⑮ $\frac{1}{3}$ or $\frac{1}{6}$

⑯ $\frac{2}{7}$ or $\frac{1}{7}$

⑰ $\frac{4}{8}$ or $\frac{5}{8}$

⑱ $\frac{2}{7}$ or $\frac{3}{7}$

⑲ $\frac{4}{8}$ or $\frac{4}{11}$

⑳ $\frac{4}{11}$ or $\frac{4}{6}$

ROUND 2 Questions 21–32: write both fractions with the lowest common denominator.

⑳ $\frac{1}{2}$ and $\frac{7}{8}$

㉑ $\frac{2}{3}$ and $\frac{6}{12}$

㉒ $\frac{7}{10}$ and $\frac{3}{6}$

㉓ $\frac{3}{5}$ and $\frac{1}{3}$

㉔ $\frac{1}{6}$ and $\frac{2}{5}$

㉕ $\frac{3}{4}$ and $\frac{1}{12}$

㉖ $\frac{3}{6}$ and $\frac{2}{3}$

㉗ $\frac{6}{12}$ and $\frac{9}{10}$

㉘ $\frac{3}{8}$ and $\frac{9}{10}$

㉙ $\frac{11}{12}$ and $\frac{4}{5}$

㉚ $\frac{6}{10}$ and $\frac{1}{2}$

㉛ $\frac{1}{3}$ and $\frac{1}{6}$

YOU DO

On your own: **Easy** Set B. Write each fraction with the denominator shown.

6 QUESTIONS

1 $\frac{1}{2} \rightarrow$ twelfths

4 $\frac{2}{5} \rightarrow$ tenths

2 $\frac{2}{3} \rightarrow$ twelfths

5 $\frac{5}{6} \rightarrow$ twelfths

3 $\frac{3}{4} \rightarrow$ twelfths

6 $\frac{3}{8} \rightarrow$ twenty-fourths



Level 2 · Building up

Medium · 4 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Order from smallest to largest.

a Order $\frac{4}{5}, \frac{3}{10}, \frac{2}{5}, \frac{7}{10}$ from smallest to largest.

Your teacher works this out. Copy every step.

b Order $\frac{1}{2}, \frac{1}{3}, \frac{7}{12}, \frac{11}{12}$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Order from smallest to largest.

4 QUESTIONS

REMEMBER Different denominators: find the LCM of the denominators.

1 $\frac{4}{5}, \frac{3}{10}, \frac{2}{5}, \frac{7}{10}$

3 $\frac{5}{8}, \frac{1}{2}, \frac{3}{4}, \frac{3}{8}$

2 $\frac{1}{2}, \frac{1}{3}, \frac{7}{12}, \frac{11}{12}$

4 $\frac{2}{3}, \frac{4}{9}, \frac{5}{6}, \frac{1}{2}$

SKILL DRILL 2 Medium basics: keep practising

30 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 30 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: which is larger? Use a common denominator.

1 $\frac{9}{12}$ or $\frac{1}{2}$

2 $\frac{1}{2}$ or $\frac{7}{8}$

3 $\frac{3}{4}$ or $\frac{1}{6}$

4 $\frac{1}{10}$ or $\frac{2}{3}$

5 $\frac{1}{4}$ or $\frac{1}{2}$

6 $\frac{8}{10}$ or $\frac{1}{12}$

7 $\frac{3}{4}$ or $\frac{1}{3}$

8 $\frac{4}{5}$ or $\frac{1}{2}$

9 $\frac{7}{10}$ or $\frac{1}{4}$

10 $\frac{9}{10}$ or $\frac{2}{3}$

11 $\frac{4}{6}$ or $\frac{10}{12}$

12 $\frac{1}{3}$ or $\frac{1}{2}$

13 $\frac{5}{12}$ or $\frac{4}{10}$

14 $\frac{4}{5}$ or $\frac{2}{3}$

15 $\frac{3}{10}$ or $\frac{2}{5}$

16 $\frac{5}{6}$ or $\frac{2}{12}$

17 $\frac{2}{8}$ or $\frac{2}{6}$

18 $\frac{6}{10}$ or $\frac{1}{2}$

19 $\frac{1}{2}$ or $\frac{8}{12}$

20 $\frac{2}{5}$ or $\frac{7}{12}$

ROUND 2 Questions 21–30: write in ascending order (smallest first).

21 $\frac{2}{4}, \frac{8}{12}, \frac{3}{4}$

22 $\frac{10}{12}, \frac{1}{4}, \frac{4}{12}$

23 $\frac{3}{6}, \frac{2}{5}, \frac{1}{8}$

24 $\frac{1}{3}, \frac{1}{2}, \frac{4}{10}$

25 $\frac{6}{8}, \frac{4}{10}, \frac{1}{3}$

26 $\frac{5}{12}, \frac{10}{12}, \frac{1}{2}$

27 $\frac{2}{5}, \frac{1}{3}, \frac{3}{8}$

28 $\frac{3}{4}, \frac{3}{5}, \frac{4}{5}$

29 $\frac{5}{6}, \frac{2}{3}, \frac{1}{2}$

30 $\frac{2}{4}, \frac{5}{6}, \frac{1}{4}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Order from smallest to largest.

a Order $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}, \frac{7}{12}$.

b Order $1\frac{1}{3}, \frac{5}{4}, \frac{11}{8}$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Order from smallest to largest.

4 QUESTIONS

REMEMBER Rewrite each fraction with that denominator.

1 $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}, \frac{7}{12}$

3 $1\frac{1}{3}, \frac{5}{4}, \frac{11}{8}, 1\frac{1}{6}$

2 $\frac{3}{5}, \frac{1}{2}, \frac{7}{10}, \frac{4}{15}$

4 Find a fraction between $\frac{1}{3}$ and $\frac{1}{2}$.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Which is larger: $\frac{4}{9}$ or $\frac{7}{9}$?

Level 2

Which is larger: $\frac{3}{4}$ or $\frac{4}{5}$?

Level 3

Order $\frac{5}{6}, \frac{2}{3}, \frac{3}{4}$.

Extension · Stretch yourself**Worked example**

Order $\frac{5}{7}$, $\frac{2}{3}$, $\frac{7}{10}$ from smallest to largest.

1. Common denominator 210
2. $\frac{150}{210}$, $\frac{140}{210}$, $\frac{147}{210}$

Answer: $\frac{2}{3}$, $\frac{7}{10}$, $\frac{5}{7}$

YOU DO

On your own: read the worked example, then try these.

- E1 Order $\frac{4}{9}$, $\frac{3}{7}$, $\frac{1}{2}$.

- E2 Find a fraction exactly halfway between $\frac{1}{4}$ and $\frac{1}{3}$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Same denominator: compare the numerators.
 2. Different denominators: find the LCM of the denominators.
 3. Rewrite each fraction with that denominator.
 4. Order the numerators.

Level 1

$\frac{3}{8}$ or $\frac{5}{8}$

1. Same denominator

Answer: $\frac{5}{8}$

Level 2

$\frac{2}{3}$ or $\frac{3}{5}$

1. $\frac{10}{15}$ vs $\frac{9}{15}$

Answer: $\frac{2}{3}$

Level 3

$\frac{1}{2}$, $\frac{1}{3}$, $\frac{7}{12}$, $\frac{11}{12}$

1. Twelfths: 6, 4, 7, 11

Answer: $\frac{1}{3}$, $\frac{1}{2}$, $\frac{7}{12}$, $\frac{11}{12}$

Extension

$\frac{5}{7}$, $\frac{2}{3}$, $\frac{7}{10}$

1. Denominator 210

Answer: $\frac{2}{3}$, $\frac{7}{10}$, $\frac{5}{7}$

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LESSON
4.03

Adding and subtracting fractions

Date _____

Learning intentions

We are learning to:

- add and subtract fractions

Success criteria

I can:

- ☐ add and subtract fractions with the same denominator
- ☐ use a common denominator when the denominators are different
- ☐ simplify the answer

Key terms Write the definition as your teacher explains it.

Numerator

Denominator

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO

With your teacher: copy what they write on the board.

Example 1

Evaluate. The denominators are the same.

a $\frac{3}{4} + \frac{1}{4}$

b $\frac{5}{7} - \frac{2}{7}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: Evaluate. Simplify if you can.

12 QUESTIONS

REMEMBER Same denominator: add or subtract the numerators.

1 $\frac{3}{4} + \frac{1}{4}$

2 $\frac{2}{9} + \frac{5}{9}$

3 $\frac{7}{10} - \frac{3}{10}$

4 $\frac{5}{8} - \frac{1}{8}$

5 $\frac{4}{5} + \frac{3}{5}$

6 $\frac{11}{12} - \frac{5}{12}$

Questions 7–12: evaluate.

7 $\frac{1}{5} + \frac{2}{5}$

8 $\frac{7}{9} - \frac{4}{9}$

9 $\frac{1}{2} + \frac{1}{4}$

10 $\frac{2}{3} - \frac{1}{6}$

11 $\frac{1}{3} + \frac{1}{4}$

12 $\frac{3}{4} - \frac{2}{3}$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: evaluate. The denominators are the same. Simplify your answer.

1 $\frac{2}{3} + \frac{1}{3}$

2 $\frac{5}{6} + \frac{3}{6}$

3 $\frac{3}{4} + \frac{1}{4}$

4 $\frac{4}{6} + \frac{2}{6}$

5 $\frac{3}{4} - \frac{1}{4}$

6 $\frac{3}{5} + \frac{4}{5}$

7 $\frac{10}{11} - \frac{2}{11}$

8 $\frac{4}{5} - \frac{2}{5}$

9 $\frac{6}{7} + \frac{5}{7}$

10 $\frac{9}{11} - \frac{7}{11}$

11 $\frac{11}{12} - \frac{10}{12}$

12 $\frac{6}{8} - \frac{4}{8}$

13 $\frac{1}{12} + \frac{4}{12}$

14 $\frac{9}{10} + \frac{8}{10}$

15 $\frac{5}{7} + \frac{6}{7}$

16 $\frac{2}{4} + \frac{1}{4}$

17 $\frac{2}{4} + \frac{3}{4}$

18 $\frac{7}{9} + \frac{4}{9}$

19 $\frac{1}{5} + \frac{2}{5}$

20 $\frac{2}{3} + \frac{2}{3}$

21 $\frac{3}{9} + \frac{3}{9}$

22 $\frac{1}{11} + \frac{2}{11}$

23 $\frac{5}{7} - \frac{2}{7}$

24 $\frac{7}{8} - \frac{5}{8}$

ROUND 2 Questions 25–40: evaluate. Change one fraction so the denominators match.

25 $\frac{2}{3} + \frac{6}{9}$

26 $\frac{1}{3} + \frac{2}{6}$

27 $\frac{1}{2} + \frac{2}{4}$

28 $\frac{1}{2} - \frac{1}{4}$

29 $\frac{2}{3} + \frac{1}{6}$

30 $\frac{4}{5} - \frac{1}{10}$

31 $\frac{2}{5} + \frac{10}{15}$

32 $\frac{4}{5} - \frac{4}{15}$

33 $\frac{3}{4} - \frac{2}{12}$

34 $\frac{1}{2} + \frac{3}{4}$

35 $\frac{1}{5} + \frac{9}{15}$

36 $\frac{1}{3} + \frac{4}{6}$

37 $\frac{2}{4} - \frac{3}{8}$

38 $\frac{1}{4} + \frac{6}{8}$

39 $\frac{3}{5} + \frac{6}{15}$

40 $\frac{2}{5} + \frac{5}{10}$

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $\frac{1}{2} + \frac{1}{4}$

4 $\frac{5}{10} - \frac{1}{3}$

2 $\frac{2}{3} - \frac{1}{6}$

5 $\frac{1}{2} - \frac{3}{8}$

3 $\frac{2}{5} + \frac{3}{10}$

6 $\frac{3}{4} + \frac{1}{8}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Use a common denominator.

a $\frac{2}{5} + \frac{3}{10}$

Your teacher works this out. Copy every step.

b $\frac{3}{4} - \frac{1}{5}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate. Use a common denominator.

6 QUESTIONS

REMEMBER Different: find the lowest common denominator.

1 $\frac{1}{2} + \frac{1}{3}$

3 $\frac{2}{3} + \frac{1}{4}$

5 $\frac{5}{6} + \frac{3}{4}$

2 $\frac{3}{4} - \frac{1}{5}$

4 $\frac{4}{5} - \frac{1}{2}$

6 $\frac{13}{18} - \frac{3}{8}$

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: evaluate. Use the lowest common denominator.

① $\frac{1}{3} + \frac{7}{10}$

② $\frac{3}{6} + \frac{1}{3}$

③ $\frac{2}{6} + \frac{1}{2}$

④ $\frac{1}{2} - \frac{1}{8}$

⑤ $\frac{7}{12} + \frac{2}{4}$

⑥ $\frac{6}{8} - \frac{1}{2}$

⑦ $\frac{9}{10} + \frac{3}{6}$

⑧ $\frac{8}{10} + \frac{2}{3}$

⑨ $\frac{2}{4} + \frac{2}{3}$

⑩ $\frac{6}{10} + \frac{5}{6}$

⑪ $\frac{4}{5} - \frac{1}{2}$

⑫ $\frac{4}{8} + \frac{1}{12}$

⑬ $\frac{1}{3} + \frac{2}{8}$

⑭ $\frac{1}{4} + \frac{3}{8}$

⑮ $\frac{4}{10} + \frac{1}{8}$

⑯ $\frac{1}{2} - \frac{2}{5}$

ROUND 2 Questions 17–28: evaluate.

⑰ $\frac{1}{2} + \frac{1}{3} - \frac{1}{8}$

⑱ $\frac{1}{3} + \frac{1}{6} - \frac{1}{6}$

⑲ $\frac{1}{4} + \frac{1}{3} - \frac{1}{12}$

⑳ $\frac{1}{2} + \frac{1}{3} - \frac{1}{12}$

㉑ $\frac{1}{3} + \frac{1}{4} - \frac{1}{12}$

㉒ $\frac{1}{2} + \frac{1}{4} - \frac{1}{12}$

㉓ $\frac{1}{3} + \frac{1}{4} - \frac{1}{6}$

㉔ $\frac{1}{2} + \frac{1}{4} - \frac{1}{8}$

㉕ $\frac{1}{4} + \frac{1}{3} - \frac{1}{8}$

㉖ $\frac{1}{2} + \frac{1}{6} - \frac{1}{12}$

㉗ $\frac{1}{3} + \frac{1}{6} - \frac{1}{8}$

㉘ $\frac{1}{4} + \frac{1}{4} - \frac{1}{12}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Use the lowest common denominator.

a $\frac{1}{2} + \frac{1}{3}$

b $\frac{5}{6} + \frac{3}{4}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Rewrite, then add or subtract.

1 $\frac{5}{6} + \frac{4}{5}$

3 $\frac{7}{8} - \frac{1}{3} - \frac{1}{6}$

2 $\frac{1}{2} + \frac{1}{3} + \frac{1}{4}$

4 Tom ate $\frac{1}{3}$ of a pizza and Sue ate $\frac{2}{5}$. What fraction is left?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{2}{7} + \frac{3}{7}$

Level 2

$\frac{1}{2} - \frac{1}{6}$

Level 3

$\frac{2}{3} + \frac{3}{5}$

Extension · Stretch yourself**Worked example**Find the missing fraction: $\frac{3}{8} + ? = \frac{5}{6}$

1. $?\ = \frac{5}{6} - \frac{3}{8}$

2. $\ = \frac{20}{24} - \frac{9}{24}$

Answer: $\frac{11}{24}$ **YOU DO**

On your own: read the worked example, then try these.

E1 Find the missing fraction: $? - \frac{2}{5} = \frac{1}{3}$

E2 Write $\frac{7}{12}$ as the sum of two different unit fractions (numerator 1).

Lesson summary

Missed the lesson? Start here.

Remember 1. Same denominator: add or subtract the numerators.
3. Rewrite, then add or subtract.

2. Different: find the lowest common denominator.
4. Simplify the answer.

Level 1

$\frac{3}{4} + \frac{1}{4}$

1. $\frac{4}{4}$

Answer: 1

Level 2

$\frac{2}{5} + \frac{3}{10}$

1. $\frac{4}{10} + \frac{3}{10}$

Answer: $\frac{7}{10}$ **Level 3**

$\frac{3}{4} - \frac{1}{5}$

1. $\frac{15}{20} - \frac{4}{20}$

Answer: $\frac{11}{20}$ **Extension**

$\frac{3}{8} + ? = \frac{5}{6}$

1. $\frac{20}{24} - \frac{9}{24}$

Answer: $\frac{11}{24}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.04

Adding and subtracting mixed numerals

Date _____

Learning intentions

We are learning to:

- add and subtract mixed numerals

Success criteria

I can:

- ☐ add the whole numbers, then the fractions
- ☐ change mixed numerals to improper fractions when subtracting needs it
- ☐ write the answer as a simplified mixed numeral

Key terms Write the definition as your teacher explains it.

Mixed numeral

Whole number

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO

With your teacher: copy what they write on the board.

Example 1

Evaluate. Add the whole numbers, then the fractions.

a $2\frac{1}{5} + 6\frac{1}{2}$

b $3\frac{3}{4} + 1\frac{1}{2}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: Evaluate.

12 QUESTIONS

REMEMBER Adding: add wholes, then add fractions.

1 $2\frac{1}{5} + 1\frac{2}{5}$

2 $3\frac{1}{4} + 2\frac{1}{4}$

3 $4\frac{5}{7} - 1\frac{2}{7}$

4 $5\frac{1}{2} + 3$

5 $6\frac{7}{9} - 2\frac{4}{9}$

6 $1\frac{1}{3} + 1\frac{2}{3}$

Questions 7–12: evaluate.

7 $2\frac{1}{3} + 1\frac{1}{3}$

8 $4\frac{3}{5} - 2\frac{1}{5}$

9 $3 - \frac{1}{4}$

10 $1\frac{1}{2} + 1\frac{1}{2}$

11 $2\frac{3}{4} + \frac{1}{2}$

12 $5\frac{1}{6} - 1\frac{5}{6}$

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: evaluate. Add (or subtract) the whole numbers, then the fractions.

① $8\frac{3}{4} - 6\frac{1}{4}$

② $8\frac{2}{3} + 4\frac{1}{3}$

③ $5\frac{1}{4} + 3\frac{3}{4}$

④ $3\frac{2}{4} + 2\frac{1}{4}$

⑤ $7\frac{8}{10} - 2\frac{6}{10}$

⑥ $5\frac{1}{4} - 3\frac{1}{4}$

⑦ $3\frac{3}{5} + 2\frac{3}{5}$

⑧ $3\frac{7}{9} + 2\frac{5}{9}$

⑨ $7\frac{1}{3} + 5\frac{1}{3}$

⑩ $3\frac{3}{5} + 1\frac{2}{5}$

⑪ $6\frac{6}{10} + 4\frac{5}{10}$

⑫ $8\frac{6}{8} - 2\frac{2}{8}$

⑬ $7\frac{1}{7} - 2\frac{1}{7}$

⑭ $8\frac{5}{6} - 6\frac{3}{6}$

⑮ $7\frac{2}{3} + 6\frac{1}{3}$

⑯ $7\frac{1}{4} - 4\frac{1}{4}$

⑰ $6\frac{3}{6} - 1\frac{2}{6}$

⑱ $5\frac{9}{10} - 4\frac{9}{10}$

⑲ $7\frac{4}{5} - 5\frac{3}{5}$

⑳ $3\frac{6}{7} + 2\frac{2}{7}$

ROUND 2 Questions 21–36: evaluate. Take the fraction from one whole.

⑳ $8 - \frac{1}{9}$

㉑ $5 - \frac{5}{6}$

㉒ $4 - \frac{3}{6}$

㉓ $9 - \frac{1}{4}$

㉔ $5 - \frac{4}{5}$

㉕ $8 - \frac{5}{8}$

㉖ $2 - \frac{9}{10}$

㉗ $5 - \frac{2}{3}$

㉘ $7 - \frac{6}{7}$

㉙ $7 - \frac{3}{5}$

㉚ $7 - \frac{2}{3}$

㉛ $7 - \frac{1}{4}$

㉜ $3 - \frac{2}{3}$

㉝ $7 - \frac{5}{9}$

㉞ $2 - \frac{1}{2}$

㉟ $3 - \frac{1}{5}$

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $10 - \frac{1}{3}$

4 $4 - 1\frac{2}{5}$

2 $5 - \frac{3}{4}$

5 $3\frac{5}{6} + \frac{1}{6}$

3 $2\frac{3}{4} + 1\frac{3}{4}$

6 $2\frac{1}{8} - \frac{3}{8}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $10 - \frac{1}{3}$

Your teacher works this out. Copy every step.

b $3\frac{2}{3} - 2\frac{1}{6}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Carry if the fraction is more than 1.

1 $2\frac{1}{5} + 6\frac{1}{2}$

3 $3\frac{3}{4} + 1\frac{1}{2}$

5 $1\frac{2}{3} + 2\frac{3}{4}$

2 $3\frac{2}{3} - 2\frac{1}{6}$

4 $4\frac{3}{4} - 2\frac{1}{3}$

6 $3\frac{5}{6} - 1\frac{1}{4}$

SKILL DRILL 2 Medium basics: keep practising

24 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: evaluate. Use a common denominator.

① $4\frac{1}{2} + 1\frac{2}{5}$

② $5\frac{1}{2} + 5\frac{3}{4}$

③ $3\frac{1}{5} + 5\frac{1}{2}$

④ $3\frac{1}{2} + 1\frac{1}{8}$

⑤ $4\frac{1}{3} + 3\frac{5}{12}$

⑥ $5\frac{9}{10} + 1\frac{2}{3}$

⑦ $6\frac{1}{8} + 1\frac{1}{3}$

⑧ $3\frac{3}{5} + 3\frac{1}{3}$

⑨ $4\frac{5}{12} + 4\frac{1}{2}$

⑩ $2\frac{5}{8} + 5\frac{1}{3}$

⑪ $2\frac{3}{4} + 4\frac{7}{8}$

⑫ $3\frac{7}{12} + 4\frac{2}{3}$

ROUND 2 Questions 13–24: evaluate. You may need to change to improper fractions.

⑬ $3\frac{3}{8} - 1\frac{7}{8}$

⑭ $8\frac{1}{5} - 1\frac{2}{5}$

⑮ $5\frac{1}{4} - 2\frac{3}{4}$

⑯ $7\frac{1}{5} - 2\frac{4}{5}$

⑰ $8\frac{1}{3} - 3\frac{2}{3}$

⑱ $6\frac{1}{3} - 2\frac{2}{3}$

⑲ $8\frac{2}{5} - 5\frac{4}{5}$

⑳ $8\frac{1}{4} - 7\frac{3}{4}$

㉑ $9\frac{3}{5} - 7\frac{4}{5}$

㉒ $9\frac{1}{3} - 1\frac{2}{3}$

㉓ $5\frac{1}{8} - 3\frac{5}{8}$

㉔ $7\frac{1}{5} - 4\frac{4}{5}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions.

a $5\frac{1}{4} - 3\frac{2}{3}$

b $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Subtracting: change to improper fractions if needed.

1 $5\frac{1}{4} - 3\frac{2}{3}$

3 $2\frac{1}{2} + 1\frac{1}{3} + 2\frac{3}{4}$

2 $3\frac{1}{2} - 1\frac{4}{5}$

4 A plank is $3\frac{1}{2}$ m long. $1\frac{3}{4}$ m is cut off. How long is the rest?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$2\frac{1}{3} + 4\frac{1}{3}$

Level 2

$6 - \frac{2}{5}$

Level 3

$4\frac{1}{3} - 2\frac{3}{4}$

Extension · Stretch yourself**Worked example**

Evaluate $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$.

1. Improper: $\frac{11}{4} - \frac{8}{5} + \frac{3}{2}$
2. Denominator 20: $\frac{55}{20} - \frac{32}{20} + \frac{30}{20}$
3. $\frac{53}{20}$

Answer: $2\frac{13}{20}$

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $5\frac{1}{3} - 2\frac{1}{2} - 1\frac{3}{4}$.

E2 Two mixed numerals add to $7\frac{1}{6}$. One is $3\frac{2}{3}$. Find the other.

Lesson summary

Missed the lesson? Start here.

Remember

1. Adding: add wholes, then add fractions.
2. Carry if the fraction is more than 1.
3. Subtracting: change to improper fractions if needed.
4. Write the answer as a mixed numeral.

Level 1

$$2\frac{1}{5} + 6\frac{1}{2}$$

1. $8 + \frac{7}{10}$

Answer: $8\frac{7}{10}$

Level 2

$$10 - \frac{1}{3}$$

1. $9\frac{3}{3} - \frac{1}{3}$

Answer: $9\frac{2}{3}$

Level 3

$$5\frac{1}{4} - 3\frac{2}{3}$$

1. $\frac{63}{12} - \frac{44}{12}$

Answer: $1\frac{7}{12}$

Extension

$$2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$$

1. $\frac{53}{20}$

Answer: $2\frac{13}{20}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.05

Fraction of a quantity

Date _____

Learning intentions

We are learning to:

- find a fraction of a quantity, including with units

Success criteria

I can:

- ☐ divide by the denominator, then multiply by the numerator
- ☐ convert units before finding the fraction
- ☐ give the answer in the unit asked for

Key terms Write the definition as your teacher explains it.**Quantity****Unit**

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate.

a $\frac{1}{4}$ of 28

b $\frac{3}{7}$ of 21

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate.

12 QUESTIONS

REMEMBER “of” means multiply.

1 $\frac{1}{4}$ of 28

2 $\frac{1}{10}$ of 35

3 $\frac{2}{5}$ of 20

4 $\frac{4}{9}$ of 27

5 $\frac{2}{3}$ of 30

6 $\frac{3}{7}$ of 21

Questions 7–12: evaluate.

7 $\frac{1}{3}$ of 18

8 $\frac{1}{5}$ of 40

9 $\frac{3}{4}$ of 20

10 $\frac{2}{3}$ of 12

11 $\frac{5}{8}$ of 32

12 $\frac{3}{10}$ of 50

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: evaluate. Divide by the denominator.

1 $\frac{1}{10}$ of 60

2 $\frac{1}{6}$ of 18

3 $\frac{1}{2}$ of 12

4 $\frac{1}{10}$ of 120

5 $\frac{1}{6}$ of 12

6 $\frac{1}{5}$ of 15

7 $\frac{1}{7}$ of 21

8 $\frac{1}{2}$ of 16

9 $\frac{1}{12}$ of 36

10 $\frac{1}{2}$ of 14

11 $\frac{1}{12}$ of 96

12 $\frac{1}{7}$ of 77

13 $\frac{1}{6}$ of 54

14 $\frac{1}{4}$ of 36

15 $\frac{1}{3}$ of 6

16 $\frac{1}{5}$ of 20

17 $\frac{1}{10}$ of 110

18 $\frac{1}{7}$ of 14

19 $\frac{1}{8}$ of 16

20 $\frac{1}{3}$ of 9

21 $\frac{1}{11}$ of 33

22 $\frac{1}{11}$ of 22

23 $\frac{1}{7}$ of 70

24 $\frac{1}{12}$ of 84

ROUND 2 Questions 25–40: evaluate. Divide by the denominator, then multiply by the numerator.

25 $\frac{5}{7}$ of 70

26 $\frac{7}{10}$ of 20

27 $\frac{9}{10}$ of 80

28 $\frac{5}{6}$ of 18

29 $\frac{3}{10}$ of 70

30 $\frac{4}{5}$ of 20

31 $\frac{4}{5}$ of 45

32 $\frac{4}{7}$ of 56

33 $\frac{3}{5}$ of 50

34 $\frac{7}{9}$ of 90

35 $\frac{3}{7}$ of 77

36 $\frac{2}{3}$ of 27

37 $\frac{2}{5}$ of 40

38 $\frac{4}{9}$ of 99

39 $\frac{4}{5}$ of 50

40 $\frac{7}{10}$ of 30

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $\frac{3}{8}$ of 64

4 $\frac{3}{4}$ of 1000

2 $\frac{5}{6}$ of 42

5 $\frac{2}{9}$ of 54

3 $\frac{7}{10}$ of 150

6 $\frac{5}{12}$ of 96



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate in the unit given.

a $\frac{2}{3}$ of 1 day (in hours)

Your teacher works this out. Copy every step.

b $\frac{4}{5}$ of 2 km (in metres)

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate in the unit shown.

6 QUESTIONS

REMEMBER Divide by the denominator.

1 $\frac{2}{3}$ of 1 day (hours)

3 $\frac{3}{4}$ of 3 h (min)

5 $\frac{1}{6}$ of 5 min (s)

2 $\frac{4}{5}$ of 2 km (m)

4 $\frac{7}{10}$ of 15 cm (mm)

6 $\frac{5}{8}$ of 2 m (cm)

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: evaluate in the unit given.

1 $\frac{19}{25}$ of 1 kg (g)

2 $\frac{1}{2}$ of 1 kg (g)

3 $\frac{3}{4}$ of a day (h)

4 $\frac{1}{20}$ of 1 kg (g)

5 $\frac{3}{5}$ of an hour (min)

6 $\frac{1}{5}$ of 1 m (cm)

7 $\frac{3}{8}$ of 1 km (m)

8 $\frac{1}{2}$ of an hour (min)

9 $\frac{2}{3}$ of a day (h)

10 $\frac{1}{2}$ of a day (h)

11 $\frac{1}{4}$ of 1 km (m)

12 $\frac{1}{4}$ of 1 kg (g)

13 $\frac{1}{10}$ of \$1 (c)

14 $\frac{4}{25}$ of \$1 (c)

15 $\frac{23}{25}$ of 1 kg (g)

ROUND 2 Questions 16–31: what fraction is the first amount of the second? Simplify.

16 24 of 30

17 4 of 20

18 10 of 24

19 10 of 15

20 2 of 12

21 12 of 60

22 8 of 15

23 13 of 24

24 8 of 20

25 36 of 40

26 5 of 12

27 3 of 15

28 11 of 24

29 8 of 12

30 16 of 20

31 8 of 30

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate in the unit given.

a $\frac{3}{4}$ of 3 hours (in minutes)

b $\frac{1}{6}$ of 5 minutes (in seconds)

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Word problems.

4 QUESTIONS

REMEMBER Multiply by the numerator.

1 $\frac{3}{5}$ of a class of 30 are girls. How many boys are there?

3 $\frac{2}{3}$ of a number is 24. Find the number.

2 A phone costs \$840. Mia has saved $\frac{5}{7}$ of the price. How much more does she need?

4 Which is more: $\frac{3}{4}$ of an hour or $\frac{7}{10}$ of an hour? By how many minutes?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{1}{5}$ of 45

Level 2

$\frac{5}{6}$ of 72

Level 3

$\frac{3}{4}$ of 2 hours (in minutes)

Extension · Stretch yourself**Worked example**

After spending $\frac{1}{4}$ of his money and then $\frac{1}{3}$ of what was left, Ali has \$30. How much did he start with?

1. After $\frac{1}{4}$: $\frac{3}{4}$ left
2. Then $\frac{2}{3}$ of $\frac{3}{4} = \frac{1}{2}$ of the money is left
3. $\frac{1}{2} = \$30$

Answer: \$60**YOU DO**

On your own: read the worked example, then try these.

- E1** $\frac{2}{5}$ of a number is 18. What is $\frac{3}{4}$ of the number?

- E2** A tank is $\frac{3}{8}$ full. After adding 60 L it is $\frac{3}{4}$ full. How much does it hold?

Lesson summary

Missed the lesson? Start here.

Remember 1. “of” means multiply.
3. Multiply by the numerator.

2. Divide by the denominator.
4. Change units first if needed.

Level 1 $\frac{3}{7}$ of 21

1. $21 \div 7 = 3$
2. 3×3

Answer: 9**Level 2** $\frac{4}{5}$ of 2 km in m

1. 2 km = 2000 m
2. $2000 \div 5 \times 4$

Answer: 1600 m**Level 3** $\frac{3}{4}$ of 3 h in min

1. $180 \div 4 \times 3$

Answer: 135 min**Extension**

Ali's money

1. $\frac{1}{2}$ left is \$30

Answer: \$60

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LESSON
4.06

Multiplying fractions

Date _____

Learning intentions

We are learning to:

- multiply fractions and mixed numerals

Success criteria

I can:

- ☐ multiply numerators and multiply denominators
- ☐ cancel common factors first
- ☐ change mixed numerals to improper fractions first

Key terms Write the definition as your teacher explains it.

Product

Cancel

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate.

a $\frac{2}{3} \times \frac{1}{5}$

b $10 \times \frac{3}{5}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate.

12 QUESTIONS

REMEMBER Change mixed numerals to improper fractions.

1 $\frac{2}{3} \times \frac{1}{5}$

2 $\frac{1}{3} \times \frac{1}{7}$

3 $\frac{3}{4} \times \frac{1}{2}$

4 $\frac{2}{5} \times \frac{2}{3}$

5 $10 \times \frac{3}{5}$

6 $\frac{5}{6} \times 12$

Questions 7–12: evaluate.

7 $\frac{1}{2} \times \frac{1}{3}$

8 $\frac{2}{5} \times \frac{3}{4}$

9 $\frac{3}{8} \times 4$

10 $\frac{5}{6} \times \frac{3}{10}$

11 $6 \times \frac{2}{3}$

12 $\frac{4}{9} \times \frac{3}{8}$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: evaluate. Multiply the numerators and the denominators. Simplify.

1 $\frac{5}{9} \times \frac{2}{8}$

2 $\frac{1}{2} \times \frac{2}{9}$

3 $\frac{1}{5} \times \frac{6}{7}$

4 $\frac{7}{8} \times \frac{4}{8}$

5 $\frac{1}{5} \times \frac{4}{5}$

6 $\frac{1}{4} \times \frac{2}{9}$

7 $\frac{1}{3} \times \frac{3}{4}$

8 $\frac{5}{7} \times \frac{1}{8}$

9 $\frac{2}{7} \times \frac{7}{8}$

10 $\frac{1}{5} \times \frac{1}{2}$

11 $\frac{4}{5} \times \frac{3}{6}$

12 $\frac{3}{5} \times \frac{1}{2}$

13 $\frac{3}{5} \times \frac{2}{3}$

14 $\frac{2}{8} \times \frac{5}{9}$

15 $\frac{8}{9} \times \frac{1}{4}$

16 $\frac{1}{3} \times \frac{1}{8}$

17 $\frac{4}{5} \times \frac{1}{3}$

18 $\frac{2}{3} \times \frac{1}{3}$

19 $\frac{3}{4} \times \frac{1}{4}$

20 $\frac{2}{4} \times \frac{2}{5}$

21 $\frac{3}{7} \times \frac{2}{5}$

22 $\frac{1}{9} \times \frac{1}{3}$

23 $\frac{1}{4} \times \frac{3}{9}$

24 $\frac{5}{7} \times \frac{2}{8}$

ROUND 2 Questions 25–40: evaluate.

25 $4 \times \frac{1}{4}$

26 $\frac{1}{3} \times 6$

27 $\frac{6}{10} \times 4$

28 $\frac{1}{5} \times 8$

29 $\frac{6}{10} \times 12$

30 $\frac{5}{7} \times 11$

31 $\frac{2}{3} \times 9$

32 $\frac{4}{6} \times 5$

33 $2 \times \frac{1}{2}$

34 $11 \times \frac{2}{4}$

35 $11 \times \frac{1}{4}$

36 $\frac{2}{10} \times 6$

37 $5 \times \frac{2}{4}$

38 $10 \times \frac{1}{7}$

39 $5 \times \frac{4}{5}$

40 $\frac{4}{9} \times 7$

YOU DO

On your own: **Easy** Set B. Evaluate. Cancel first.

6 QUESTIONS

1 $\frac{4}{7} \times \frac{5}{8}$

4 $\frac{9}{16} \times \frac{2}{45}$

2 $\frac{3}{10} \times \frac{5}{9}$

5 $\frac{4}{15} \times \frac{5}{8}$

3 $\frac{6}{7} \times \frac{7}{12}$

6 $\frac{2}{3} \times \frac{9}{10}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Cancel first.

a $\frac{4}{7} \times \frac{5}{8}$

Your teacher works this out. Copy every step.

b $\frac{9}{16} \times \frac{2}{45}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Cancel common factors (top with bottom).

1 $1\frac{1}{2} \times \frac{1}{2}$

3 $1\frac{1}{2} \times 1\frac{1}{2}$

5 $\frac{3}{4} \times 2\frac{1}{3}$

2 $2\frac{1}{5} \times \frac{1}{2}$

4 $2\frac{2}{3} \times \frac{1}{2}$

6 $3\frac{3}{5} \times \frac{3}{8}$

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: evaluate. Cancel first.

① $\frac{6}{12} \times \frac{3}{10}$

② $\frac{20}{35} \times \frac{5}{16}$

③ $\frac{9}{28} \times \frac{8}{12}$

④ $\frac{20}{30} \times \frac{24}{35}$

⑤ $\frac{12}{30} \times \frac{5}{21}$

⑥ $\frac{12}{18} \times \frac{18}{20}$

⑦ $\frac{4}{20} \times \frac{4}{12}$

⑧ $\frac{5}{20} \times \frac{10}{15}$

⑨ $\frac{15}{35} \times \frac{5}{12}$

⑩ $\frac{12}{16} \times \frac{4}{42}$

⑪ $\frac{4}{42} \times \frac{12}{14}$

⑫ $\frac{12}{16} \times \frac{4}{18}$

⑬ $\frac{6}{12} \times \frac{8}{12}$

⑭ $\frac{6}{18} \times \frac{9}{15}$

⑮ $\frac{16}{42} \times \frac{6}{20}$

⑯ $\frac{6}{10} \times \frac{8}{36}$

ROUND 2 Questions 17–28: evaluate. Change to improper fractions first.

⑰ $3\frac{1}{2} \times 3\frac{2}{3}$

⑱ $1\frac{1}{3} \times 3\frac{4}{5}$

⑲ $2\frac{1}{3} \times 1\frac{2}{3}$

⑳ $1\frac{1}{2} \times 2\frac{3}{5}$

㉑ $1\frac{1}{2} \times 2\frac{2}{3}$

㉒ $1\frac{2}{3} \times 1\frac{2}{3}$

㉓ $3\frac{1}{3} \times 3\frac{1}{3}$

㉔ $3\frac{2}{4} \times 3\frac{2}{3}$

㉕ $2\frac{1}{3} \times 3\frac{1}{2}$

㉖ $1\frac{1}{2} \times 2\frac{1}{2}$

㉗ $3\frac{2}{3} \times 1\frac{2}{4}$

㉘ $3\frac{2}{3} \times 2\frac{2}{3}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions first.

a $1\frac{1}{2} \times \frac{1}{2}$

b $2\frac{1}{3} \times 4\frac{2}{3}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Multiply tops, multiply bottoms.

1 $2\frac{1}{3} \times 4\frac{2}{3}$

3 $2\frac{1}{2} \times 1\frac{2}{5} \times \frac{1}{3}$

2 $3\frac{2}{4} \times \frac{2}{3}$

4 A recipe needs $2\frac{3}{4}$ cups of flour. How much for $\frac{2}{3}$ of the recipe?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{3}{5} \times \frac{1}{4}$

Level 2

$\frac{5}{9} \times \frac{3}{10}$

Level 3

$1\frac{1}{4} \times 2\frac{2}{3}$

Extension · Stretch yourself**Worked example**Evaluate $\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6}$.

1. Cancel: 3, 4 and 5 cancel
2. Leaves $\frac{2}{6}$

Answer: $\frac{1}{3}$ **YOU DO**

On your own: read the worked example, then try these.

E1 Evaluate $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{9}{10}$.**E2** A rectangle is $3\frac{1}{2}$ m by $2\frac{2}{3}$ m. Find its area.**Lesson summary**

Missed the lesson? Start here.

- Remember**
1. Change mixed numerals to improper fractions.
 2. Cancel common factors (top with bottom).
 3. Multiply tops, multiply bottoms.
 4. Write as a simplified mixed numeral.

Level 1

$$\frac{2}{3} \times \frac{1}{5}$$

1. 2×1 and 3×5

Answer: $\frac{2}{15}$ **Level 2**

$$\frac{4}{7} \times \frac{5}{8}$$

1. Cancel 4 and 8

Answer: $\frac{5}{14}$ **Level 3**

$$2\frac{1}{3} \times 4\frac{2}{3}$$

$$1. \frac{7}{3} \times \frac{14}{3}$$

Answer: $10\frac{8}{9}$ **Extension**

Telescoping product

1. Cancel

Answer: $\frac{1}{3}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.07

Dividing fractions

Date _____

Learning intentions

We are learning to:

- divide fractions and mixed numerals

Success criteria

I can:

- ☐ find the **reciprocal** of a fraction
- ☐ divide by multiplying by the reciprocal
- ☐ change mixed numerals to improper fractions first

Key terms Write the definition as your teacher explains it.**Reciprocal****Quotient**

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Write the reciprocal, then divide.**a** Write the reciprocal of $\frac{3}{4}$, 5 and $2\frac{1}{2}$.**b** $10 \div \frac{1}{5}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write the reciprocal.

12 QUESTIONS

REMEMBER Change mixed numerals to improper fractions.

1 $\frac{3}{4}$

2 $\frac{2}{5}$

3 7

4 $\frac{1}{9}$

5 $\frac{8}{3}$

6 $1\frac{1}{2}$

Questions 7–12: evaluate.

7 $6 \div \frac{1}{2}$

8 $\frac{1}{3} \div 2$

9 $\frac{3}{4} \div \frac{1}{4}$

10 $\frac{2}{5} \div \frac{4}{5}$

11 $\frac{5}{6} \div \frac{1}{3}$

12 $4 \div \frac{2}{3}$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write the reciprocal.

1 $\frac{10}{2}$

2 5

3 $\frac{4}{3}$

4 $\frac{5}{10}$

5 $\frac{4}{7}$

6 4

7 8

8 $\frac{10}{4}$

9 $\frac{1}{6}$

10 $\frac{7}{8}$

11 3

12 6

13 $\frac{5}{7}$

14 $\frac{8}{2}$

15 $\frac{4}{6}$

16 $\frac{3}{12}$

17 $\frac{1}{3}$

18 12

19 $\frac{3}{2}$

20 $\frac{10}{7}$

21 $\frac{11}{12}$

22 9

23 $\frac{8}{4}$

24 $\frac{7}{11}$

ROUND 2 Questions 25–40: evaluate. How many parts fit into the whole number?

25 $6 \div \frac{1}{2}$

26 $2 \div \frac{1}{7}$

27 $2 \div \frac{1}{6}$

28 $6 \div \frac{1}{6}$

29 $8 \div \frac{1}{5}$

30 $5 \div \frac{1}{3}$

31 $6 \div \frac{1}{7}$

32 $7 \div \frac{1}{7}$

33 $4 \div \frac{1}{5}$

34 $4 \div \frac{1}{3}$

35 $5 \div \frac{1}{5}$

36 $9 \div \frac{1}{2}$

37 $3 \div \frac{1}{8}$

38 $2 \div \frac{1}{8}$

39 $7 \div \frac{1}{3}$

40 $8 \div \frac{1}{6}$

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $10 \div \frac{1}{5}$

4 $6 \div \frac{2}{3}$

2 $\frac{1}{2} \div 3$

5 $\frac{4}{5} \div 2$

3 $\frac{1}{2} \div \frac{1}{3}$

6 $3 \div \frac{1}{4}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Multiply by the reciprocal.

a $\frac{1}{2} \div 3$

Your teacher works this out. Copy every step.

b $\frac{3}{4} \div \frac{3}{8}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Keep the first fraction.

1 $\frac{3}{4} \div \frac{3}{8}$

3 $\frac{4}{7} \div \frac{5}{8}$

5 $\frac{2}{3} \div \frac{5}{6}$

2 $\frac{3}{5} \div \frac{1}{10}$

4 $\frac{17}{20} \div \frac{3}{5}$

6 $\frac{5}{9} \div \frac{2}{3}$

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: evaluate. Multiply by the reciprocal.

① $\frac{4}{6} \div \frac{1}{3}$

② $\frac{1}{3} \div \frac{1}{2}$

③ $\frac{1}{3} \div \frac{6}{7}$

④ $\frac{2}{5} \div \frac{4}{5}$

⑤ $\frac{2}{4} \div \frac{1}{5}$

⑥ $\frac{5}{6} \div \frac{1}{4}$

⑦ $\frac{1}{3} \div \frac{2}{3}$

⑧ $\frac{1}{2} \div \frac{3}{6}$

⑨ $\frac{1}{2} \div \frac{1}{5}$

⑩ $\frac{8}{9} \div \frac{2}{6}$

⑪ $\frac{3}{8} \div \frac{3}{9}$

⑫ $\frac{1}{2} \div \frac{2}{9}$

⑬ $\frac{2}{6} \div \frac{2}{3}$

⑭ $\frac{1}{7} \div \frac{3}{5}$

⑮ $\frac{5}{7} \div \frac{6}{8}$

⑯ $\frac{3}{7} \div \frac{2}{3}$

ROUND 2 Questions 17–28: evaluate. Change to improper fractions first.

⑰ $3\frac{1}{2} \div 2\frac{1}{2}$

⑱ $2\frac{1}{2} \div 1\frac{2}{5}$

⑲ $1\frac{2}{4} \div 1\frac{2}{4}$

⑳ $2\frac{1}{2} \div 1\frac{2}{3}$

㉑ $2\frac{3}{5} \div 2\frac{1}{2}$

㉒ $3\frac{1}{5} \div 1\frac{1}{2}$

㉓ $3\frac{1}{2} \div 2\frac{2}{4}$

㉔ $4\frac{2}{3} \div 1\frac{1}{5}$

㉕ $2\frac{3}{5} \div 1\frac{1}{4}$

㉖ $4\frac{1}{2} \div 1\frac{2}{3}$

㉗ $2\frac{1}{3} \div 2\frac{1}{3}$

㉘ $4\frac{3}{5} \div 2\frac{1}{3}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions first.

a $2\frac{1}{4} \div 1\frac{1}{8}$

b $1\frac{7}{8} \div 1\frac{1}{4}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Change $\div$ to $\times$ and flip the second fraction.

1 $1\frac{1}{2} \div \frac{1}{2}$

3 $1\frac{7}{8} \div 1\frac{1}{4}$

2 $2\frac{1}{4} \div 1\frac{1}{8}$

4 How many $\frac{3}{4}$ m pieces can be cut from a $7\frac{1}{2}$ m rope?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write the reciprocal of $\frac{5}{8}$.

Level 2

$\frac{2}{3} \div \frac{1}{6}$

Level 3

$3\frac{1}{2} \div 1\frac{3}{4}$

Extension · Stretch yourself**Worked example**

A jug holds $2\frac{1}{4}$ L. How many $\frac{3}{10}$ L glasses can be filled?

$$1. \frac{9}{4} \div \frac{3}{10} = \frac{9}{4} \times \frac{10}{3}$$

$$2. = \frac{90}{12} = 7\frac{1}{2}$$

Answer: 7 full glasses

YOU DO

On your own: read the worked example, then try these.

E1 $\frac{2}{3}$ of a number is $\frac{4}{9}$. Find the number.

E2 Evaluate $(\frac{1}{2} + \frac{1}{3}) \div (\frac{1}{2} - \frac{1}{3})$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Change mixed numerals to improper fractions.
2. Keep the first fraction.
3. Change $\div$ to $\times$ and flip the second fraction.
4. Multiply and simplify.

Level 1

$$10 \div \frac{1}{5}$$

$$1. 10 \times 5$$

Answer: 50

Level 2

$$\frac{3}{4} \div \frac{3}{8}$$

$$1. \frac{3}{4} \times \frac{8}{3}$$

Answer: 2

Level 3

$$2\frac{1}{4} \div 1\frac{1}{8}$$

$$1. \frac{9}{4} \times \frac{8}{9}$$

Answer: 2

Extension

Glasses from a jug

$$1. 7\frac{1}{2}$$

Answer: 7 glasses

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.08

Fraction problems

Date _____

Learning intentions

We are learning to:

- solve problems involving fractions

Success criteria

I can:

- ☐ decide which operation the problem needs
- ☐ find the “rest” by subtracting fractions from 1
- ☐ check the answer makes sense

Key terms Write the definition as your teacher explains it.

Remainder

Total

Notes

Level 1 · Getting started

Easy · 15 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Read the problem, then answer.

- a** A punch has 2 L ginger ale, $1\frac{1}{2}$ L orange juice, $2\frac{3}{4}$ L lemonade and $\frac{1}{3}$ L raspberry juice. How many litres of punch?

- b** How much more lemonade than orange juice is in the punch?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Solve.

6 QUESTIONS

REMEMBER Read the problem twice. Underline the numbers.

- ① Sam ate $\frac{1}{4}$ of a cake and Ben ate $\frac{3}{8}$. How much did they eat together?

- ③ $\frac{3}{5}$ of 40 students walk to school. How many walk?

- ⑤ A pizza has 8 slices. Lee eats 3. What fraction is left?

- ② A tank is $\frac{5}{6}$ full. $\frac{1}{3}$ is used. How full is it now?

- ④ What fraction of an hour is 20 minutes?

- ⑥ How many $\frac{1}{4}$ -hour lessons fit in 3 hours?

YOU DO On your own: **Easy** Set A2. Quick drill: what fraction is left?

6 QUESTIONS

- ① $\frac{1}{4}$ is eaten

- ② $\frac{2}{5}$ is spent

- ③ $\frac{1}{3}$ and $\frac{1}{3}$ are used

- ④ $\frac{1}{2}$ and $\frac{1}{4}$ are sold

- ⑤ $\frac{3}{8}$ is poured out

- ⑥ $\frac{1}{6}$ and $\frac{1}{3}$ are given away

YOU DO On your own: **Easy** Set B. Solve.

3 QUESTIONS

- ① A ribbon is $4\frac{1}{2}$ m. $1\frac{2}{3}$ m is cut off. How much is left?

- ② A bag of flour weighs $\frac{3}{4}$ kg. What do 6 bags weigh?

- ③ $\frac{1}{3}$ of the class have a dog and $\frac{1}{4}$ have a cat. What fraction have neither (no one has both)?

SKILL DRILL 1 Easy basics: practise until it is automatic

30 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 30 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–18: what fraction is left?1 $\frac{3}{5}$ of a garden is planted
_____2 $\frac{6}{8}$ of a pizza is eaten
_____3 $\frac{8}{12}$ of a pizza is eaten
_____4 $\frac{5}{6}$ of a class is away
_____5 $\frac{2}{10}$ of a class is away
_____6 $\frac{2}{6}$ of a class is away
_____7 $\frac{2}{3}$ of a pizza is eaten
_____8 $\frac{6}{10}$ of a pizza is eaten
_____9 $\frac{2}{5}$ of a book is read
_____10 $\frac{4}{6}$ of a bottle of water is drunk
_____11 $\frac{1}{4}$ of a pizza is eaten
_____12 $\frac{8}{12}$ of a garden is planted
_____13 $\frac{1}{3}$ of a tank of petrol is used
_____14 $\frac{4}{5}$ of a cake is sold
_____15 $\frac{8}{12}$ of a bottle of water is drunk
_____16 $\frac{1}{12}$ of a class is away
_____17 $\frac{5}{8}$ of a cake is sold
_____18 $\frac{1}{3}$ of a class is away
_____**ROUND 2** Questions 19–30: find the amount.19 $\frac{1}{2}$ of 12 kilograms
_____20 $\frac{1}{5}$ of \$15
_____21 $\frac{1}{5}$ of \$20
_____22 $\frac{2}{5}$ of 45 kilograms
_____23 $\frac{1}{10}$ of 100 minutes
_____24 $\frac{1}{2}$ of \$20
_____25 $\frac{1}{2}$ of 16 minutes
_____26 $\frac{1}{2}$ of 18 kilograms
_____27 $\frac{3}{4}$ of 28 minutes
_____28 $\frac{7}{10}$ of \$20
_____29 $\frac{2}{3}$ of \$27
_____30 $\frac{5}{6}$ of \$60

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Read the problem, then answer.

- a** A farmer has 24 goats. $\frac{1}{4}$ are black and $\frac{2}{3}$ are white. What fraction are neither?

Your teacher works this out. Copy every step.

- b** The rest of the goats are split equally between brown and spotted. How many are spotted?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Solve.

6 QUESTIONS

REMEMBER “Of” → multiply. “Altogether” → add.

- 1** A punch uses $1\frac{1}{2}$ L juice, $2\frac{3}{4}$ L soda and $\frac{1}{3}$ L syrup. Find the total.

- 3** A 60 L tank is $\frac{3}{5}$ full. How many litres to fill it?

- 5** How many $\frac{2}{3}$ -cup serves are in 8 cups?

- 2** 24 goats: $\frac{1}{4}$ black, $\frac{2}{3}$ white, the rest split equally brown and spotted. How many spotted?

- 4** Mia spends $\frac{1}{3}$ of \$90, then $\frac{1}{2}$ of what is left. How much is left?

- 6** A car uses $\frac{3}{4}$ of a tank for 360 km. How far on a full tank?

SKILL DRILL 2 Medium basics: keep practising

16 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 16 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–10: how much is left?

- 1 Sam had \$24 and spent $\frac{3}{4}$ of it. How much is left? _____

- 2 Sam had \$45 and spent $\frac{2}{3}$ of it. How much is left? _____

- 3 Sam had \$40 and spent $\frac{3}{5}$ of it. How much is left? _____

- 4 Sam had \$95 and spent $\frac{2}{5}$ of it. How much is left? _____

- 5 Sam had \$88 and spent $\frac{5}{8}$ of it. How much is left? _____

- 6 Sam had \$20 and spent $\frac{4}{5}$ of it. How much is left? _____

- 7 Sam had \$72 and spent $\frac{5}{8}$ of it. How much is left? _____

- 8 Sam had \$18 and spent $\frac{1}{3}$ of it. How much is left? _____

- 9 Sam had \$50 and spent $\frac{1}{5}$ of it. How much is left? _____

- 10 Sam had \$42 and spent $\frac{1}{3}$ of it. How much is left? _____

ROUND 2 Questions 11–16: two steps: find each part, then answer.

- 11 60 people: $\frac{1}{2}$ walk, $\frac{1}{5}$ ride. The rest take the bus.
How many take the bus?

- 12 120 people: $\frac{1}{4}$ walk, $\frac{1}{6}$ ride. The rest take the bus.
How many take the bus?

- 13 20 people: $\frac{1}{2}$ walk, $\frac{1}{5}$ ride. The rest take the bus.
How many take the bus?

- 14 24 people: $\frac{1}{3}$ walk, $\frac{1}{4}$ ride. The rest take the bus.
How many take the bus?

- 15 144 people: $\frac{1}{4}$ walk, $\frac{1}{6}$ ride. The rest take the bus.
How many take the bus?

- 16 90 people: $\frac{1}{3}$ walk, $\frac{1}{6}$ ride. The rest take the bus.
How many take the bus?



Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.**Example 3** Read the problem, then answer.

- a** Jo reads $\frac{2}{5}$ of a book on Monday and $\frac{1}{3}$ on Tuesday. What fraction is left?

Your teacher works this out. Copy every step.

- b** The book has 240 pages. How many pages are left?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Harder problems.

4 QUESTIONS

REMEMBER “Left” or “the rest” → subtract from 1 (the whole).

- 1** $\frac{3}{8}$ of a school's 480 students catch a bus and $\frac{1}{3}$ of the rest walk. How many walk?

- 3** After giving away $\frac{2}{7}$ of her cards, Zoe has 45 left. How many did she start with?

- 2** A rope is cut into pieces of $\frac{2}{5}$ m. There are 12 pieces and $\frac{1}{10}$ m left over. How long was the rope?

- 4** Is $\frac{2}{3}$ of $\frac{3}{4}$ of 80 bigger than $\frac{1}{2}$ of 80? Explain.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

A cyclist rides $\frac{1}{3}$ of a trip on day 1 and $\frac{1}{4}$ of the trip on day 2. She has 50 km left. How long is the trip?

1. $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$
2. $\frac{5}{12}$ of the trip is 50 km
3. $\frac{1}{12}$ is 10 km

Answer: 120 km

YOU DO

On your own: read the worked example, then try these.

- E1 $\frac{1}{4}$ of a class plays soccer, $\frac{1}{3}$ plays netball and the other 10 play neither. How many in the class?

- E2 A bottle is $\frac{2}{3}$ full. After pouring out 150 mL it is $\frac{1}{6}$ full. How much does it hold?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is $\frac{2}{5} + \frac{1}{3}$ of a whole?

Level 2

 $\frac{3}{4}$ of 36 students passed. How many failed?

Level 3

After spending $\frac{3}{5}$ of his money, Kai has \$28. How much did he start with?

Lesson summary

Missed the lesson? Start here.

Remember

1. Read the problem twice. Underline the numbers.
2. “Of” → multiply. “Altogether” → add.
3. “Left” or “the rest” → subtract from 1 (the whole).
4. Answer the question with units.

Level 1

Punch total

1. $2 + 1\frac{1}{2} + 2\frac{3}{4} + \frac{1}{3}$

Answer: $6\frac{7}{12}$ L

Level 2

Goats: neither black nor white

1. $1 - \frac{1}{4} - \frac{2}{3}$

Answer: $\frac{1}{12}$

Level 3

Spotted goats

1. $\frac{1}{12}$ of 24 = 2
2. $2 \div 2$

Answer: 1 goat

Extension

Cyclist

1. $\frac{5}{12} = 50$ km

Answer: 120 km

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.09

Fractions on a calculator

Date _____

Learning intentions

We are learning to:

- use a calculator to work with fractions

Success criteria

I can:

- ☐ enter fractions and mixed numerals with the fraction key
- ☐ convert between improper fractions, mixed numerals and decimals
- ☐ check calculator answers with an estimate

Key terms Write the definition as your teacher explains it.

Fraction key

S ⇌ D key

Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the fraction key on your calculator.

a Enter $\frac{3}{4} + \frac{1}{6}$.

b Enter $3\frac{1}{2} \times \frac{2}{5}$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Use a calculator.

10 QUESTIONS

REMEMBER Use the fraction key to enter a fraction.

1 $\frac{3}{4} + \frac{1}{6}$

2 $\frac{5}{6} + \frac{4}{5}$

3 $\frac{7}{9} - \frac{1}{4}$

4 $\frac{13}{18} - \frac{3}{8}$

5 $2\frac{3}{4} + 1\frac{5}{6}$

6 $5\frac{1}{2} - 2\frac{2}{3}$

Questions 7–10: use your calculator.

7 $\frac{5}{12} + \frac{7}{18}$

8 $\frac{11}{15} - \frac{3}{10}$

9 $\frac{7}{12} \times \frac{9}{14}$

10 $\frac{5}{9} \div \frac{10}{27}$

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: use the fraction key on your calculator.

① $\frac{9}{18} - \frac{4}{11}$

② $\frac{5}{6} - \frac{1}{8}$

③ $\frac{3}{8} + \frac{3}{12}$

④ $\frac{11}{18} + \frac{9}{13}$

⑤ $\frac{10}{19} - \frac{5}{13}$

⑥ $\frac{3}{8} + \frac{1}{5}$

⑦ $\frac{6}{9} + \frac{1}{5}$

⑧ $\frac{5}{7} - \frac{1}{12}$

⑨ $\frac{1}{15} + \frac{11}{16}$

⑩ $\frac{6}{20} + \frac{2}{19}$

⑪ $\frac{4}{6} + \frac{6}{7}$

⑫ $\frac{2}{19} + \frac{1}{9}$

⑬ $\frac{8}{13} - \frac{5}{20}$

⑭ $\frac{10}{12} + \frac{3}{18}$

⑮ $\frac{3}{6} + \frac{8}{14}$

⑯ $\frac{11}{15} - \frac{4}{9}$

⑰ $\frac{14}{18} + \frac{8}{13}$

⑱ $\frac{2}{5} - \frac{1}{7}$

⑲ $\frac{1}{5} + \frac{4}{9}$

⑳ $\frac{3}{14} + \frac{11}{17}$

ROUND 2 Questions 21–36: use your calculator.

⑳ $\frac{2}{7} \times \frac{2}{5}$

㉑ $\frac{11}{14} \div \frac{9}{11}$

㉒ $\frac{2}{5} \times \frac{8}{14}$

㉓ $\frac{18}{19} \times \frac{15}{17}$

㉔ $\frac{9}{11} \div \frac{10}{15}$

㉕ $\frac{3}{14} \times \frac{3}{8}$

㉖ $\frac{5}{10} \div \frac{4}{9}$

㉗ $\frac{2}{11} \div \frac{3}{11}$

㉘ $\frac{19}{20} \div \frac{14}{19}$

㉙ $\frac{6}{17} \times \frac{9}{11}$

㉚ $\frac{2}{6} \times \frac{7}{13}$

㉛ $\frac{1}{5} \times \frac{2}{15}$

㉜ $\frac{3}{20} \div \frac{14}{16}$

㉝ $\frac{6}{13} \times \frac{2}{7}$

㉞ $\frac{3}{17} \div \frac{16}{17}$

㉟ $\frac{8}{10} \times \frac{8}{18}$

YOU DO

On your own: **Easy** Set B. Use a calculator.

6 QUESTIONS

1 $\frac{9}{16} \times \frac{2}{45}$

4 $2\frac{3}{4} \div 1\frac{1}{5}$

2 $\frac{17}{20} \div \frac{3}{5}$

5 $\frac{5}{8}$ of 360

3 $2\frac{1}{3} \times 4\frac{2}{3}$

6 $\frac{7}{12} \times 48$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use your calculator to convert and evaluate.

- a Convert $\frac{17}{5}$ to a mixed numeral and to a decimal.

Your teacher works this out. Copy every step.

- b Evaluate $\frac{13}{18} - \frac{3}{8}$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Convert with a calculator.

6 QUESTIONS

REMEMBER Use SHIFT for a mixed numeral.

1 $\frac{17}{5}$ to a mixed numeral

3 $2\frac{3}{8}$ to an improper fraction

5 0.625 to a fraction

2 $\frac{45}{8}$ to a mixed numeral

4 $\frac{7}{20}$ to a decimal

6 $\frac{5}{16}$ to a decimal

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: use your calculator. Use brackets.

1 $(\frac{3}{4} + \frac{3}{4}) \times \frac{4}{9}$

2 $(\frac{6}{8} + \frac{1}{2}) \times \frac{3}{9}$

3 $(\frac{2}{3} + \frac{6}{8}) \times \frac{5}{6}$

4 $(\frac{1}{2} + \frac{2}{7}) \times \frac{5}{7}$

5 $(\frac{1}{7} + \frac{1}{4}) \times \frac{4}{7}$

6 $(\frac{1}{7} + \frac{2}{5}) \times \frac{3}{8}$

7 $(\frac{5}{7} + \frac{3}{7}) \times \frac{1}{3}$

8 $(\frac{5}{6} + \frac{1}{2}) \times \frac{2}{8}$

9 $(\frac{3}{8} + \frac{4}{8}) \times \frac{5}{7}$

10 $(\frac{1}{4} + \frac{7}{9}) \times \frac{1}{4}$

11 $(\frac{4}{7} + \frac{1}{4}) \times \frac{1}{2}$

12 $(\frac{4}{8} + \frac{2}{3}) \times \frac{7}{9}$

ROUND 2 Questions 13–28: use your calculator to write each as a mixed numeral.

13 $\frac{96}{11}$

14 $\frac{11}{3}$

15 $\frac{18}{13}$

16 $\frac{79}{12}$

17 $\frac{28}{15}$

18 $\frac{124}{15}$

19 $\frac{41}{9}$

20 $\frac{30}{11}$

21 $\frac{27}{14}$

22 $\frac{16}{11}$

23 $\frac{41}{15}$

24 $\frac{13}{8}$

25 $\frac{13}{6}$

26 $\frac{57}{10}$

27 $\frac{23}{3}$

28 $\frac{50}{9}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use your calculator. Use brackets.

a Evaluate $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$.

Your teacher works this out. Copy every step.

b Evaluate $2\frac{1}{4}^2$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use a calculator.

4 QUESTIONS

REMEMBER S $\leftrightarrow$ D switches between fraction and decimal.

1 $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$

3 $(\frac{3}{5} + \frac{1}{4}) \times 2\frac{2}{3}$

2 $2\frac{1}{4}^2 + \frac{1}{2}$

4 Estimate $3\frac{7}{8} \times 2\frac{1}{10}$ first, then use a calculator.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{2}{7} + \frac{3}{5}$

Level 2

$3\frac{4}{5} \times \frac{5}{6}$

Level 3

$2\frac{5}{6} \div \frac{2}{3} - \frac{1}{4}$

Extension · Stretch yourself**Worked example**

Find the fraction halfway between $\frac{5}{9}$ and $\frac{7}{11}$.

1. Add and divide by 2

2. $(\frac{5}{9} + \frac{7}{11}) \div 2$

Answer: $\frac{59}{99}$

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}$. What do you notice?

E2 Which is closer to 1: $\frac{11}{13}$ or $\frac{13}{15}$?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Use the fraction key to enter a fraction.
 2. Use SHIFT for a mixed numeral.
 3. S $\leftrightarrow$ D switches between fraction and decimal.
 4. Estimate first to check the answer.

Level 1

$$\frac{3}{4} + \frac{1}{6}$$

1. Enter with the fraction key

Answer: $\frac{11}{12}$

Level 2

$$\frac{17}{5}$$

1. Convert to mixed

Answer: $3\frac{2}{5} = 3.4$

Level 3

$$4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$$

1. Use brackets if unsure

Answer: $12\frac{1}{4}$

Extension

Halfway

1. Add and halve

Answer: $\frac{59}{99}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.10

Percentages, fractions and decimals

Date _____

Learning intentions

We are learning to:

- convert between percentages, fractions and decimals

Success criteria

I can:

- ☐ write a percentage as a fraction out of 100 and simplify
- ☐ write a percentage as a decimal by dividing by 100
- ☐ write a fraction or decimal as a percentage by multiplying by 100%

Key terms Write the definition as your teacher explains it.

Percentage

Per cent

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write as a fraction and a decimal.

a Write 13% as a fraction and a decimal.

Your teacher works this out. Copy every step.

b Write 8% as a fraction and a decimal.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write each percentage as a fraction in simplest form.

12 QUESTIONS

REMEMBER % → fraction: write over 100, simplify.

1 13%

2 8%

3 50%

4 75%

5 35%

6 120%

Questions 7–12: complete the conversion.

7 25% = ? (fraction)

8 0.6 = ?%

9 $\frac{1}{5}$ = ?%

10 40% = ? (decimal)

11 $\frac{3}{4}$ = ?%

12 0.07 = ?%

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: write each percentage as a fraction in simplest form.

1 25%

2 75%

3 35%

4 8%

5 45%

6 1%

7 80%

8 15%

9 40%

10 12%

11 90%

12 60%

13 20%

14 10%

15 4%

16 70%

17 2%

18 5%

19 30%

20 50%

ROUND 2 Questions 21–36: write each percentage as a decimal.

21 99%

22 111%

23 34%

24 22%

25 62%

26 68%

27 110%

28 10%

29 14%

30 47%

31 35%

32 59%

33 76%

34 117%

35 46%

36 118%

YOU DO

On your own: **Easy** Set B. Write each percentage as a decimal.

6 QUESTIONS

1 13%

4 150%

2 8%

5 2.5%

3 45%

6 $7\frac{1}{2}\%$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write as a fraction and a decimal, then as a percentage.

a Write $7\frac{1}{2}\%$ as a fraction and a decimal.

Your teacher works this out. Copy every step.

b Write $\frac{7}{10}$ as a percentage.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Write each as a percentage.

6 QUESTIONS

REMEMBER % → decimal: ÷ 100 (move 2 places left).

1 $\frac{7}{10}$

4 0.6

2 $\frac{23}{25}$

5 0.045

3 $\frac{3}{4}$

6 1.3

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: write each fraction as a percentage.

1 $\frac{3}{5}$

2 $\frac{8}{25}$

3 $\frac{7}{10}$

4 $\frac{17}{50}$

5 $\frac{4}{5}$

6 $\frac{9}{25}$

7 $\frac{1}{2}$

8 $\frac{1}{10}$

9 $\frac{1}{4}$

10 $\frac{2}{5}$

11 $\frac{13}{20}$

12 $\frac{23}{25}$

13 $\frac{4}{25}$

14 $\frac{13}{50}$

15 $\frac{19}{25}$

16 $\frac{21}{50}$

ROUND 2 Questions 17–31: write each decimal as a percentage and as a fraction.

17 0.3

18 0.4

19 0.2

20 0.12

21 0.5

22 0.75

23 0.25

24 0.05

25 0.02

26 0.1

27 0.04

28 0.8

29 0.7

30 0.9

31 0.45

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write as a percentage.

a Write $\frac{23}{25}$ as a percentage.

Your teacher works this out. Copy every step.

b Write 0.045 as a percentage.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Complete the conversions.

4 QUESTIONS

REMEMBER Fraction or decimal $\rightarrow$ %: $\times 100\%$.

1 Write $\frac{3}{8}$ as a decimal and a percentage.

3 Write $12\frac{1}{2}\%$ as a fraction.

2 Write $\frac{1}{3}$ as a percentage.

4 Order from smallest to largest: 0.4, 38%, $\frac{2}{5}$, $\frac{3}{8}$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 25% as a fraction.

Level 2

Write 0.07 as a percentage.

Level 3

Write $\frac{3}{16}$ as a percentage.

Extension · Stretch yourself

Worked example

Write $0.\overline{3}$ ($0.333\dots$) as a percentage and as a fraction.

- $0.333\dots \times 100\% = 33.3\dots\%$
- $= 33\frac{1}{3}\%$
- As a fraction: $\frac{1}{3}$

Answer: $33\frac{1}{3}\% = \frac{1}{3}$

YOU DO

On your own: read the worked example, then try these.

- E1 Write $\frac{2}{3}$ as a percentage.

- E2 Write $12\frac{1}{3}\%$ as a fraction in simplest form.

Lesson summary

Missed the lesson? Start here.

- Remember**
- $\% \rightarrow$ fraction: write over 100, simplify.
 - Fraction or decimal $\rightarrow \%$: $\times 100\%$.

- $\% \rightarrow$ decimal: $\div 100$ (move 2 places left).
- Learn: $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{1}{5} = 20\%$, $\frac{1}{10} = 10\%$.

Level 1

13%

1. $\frac{13}{100}$

Answer: $\frac{13}{100} = 0.13$

Level 2

$7\frac{1}{2}\%$

1. $\frac{15}{2} \div 100 = \frac{15}{200}$

Answer: $\frac{3}{40} = 0.075$

Level 3

$\frac{23}{25}$

1. $\frac{92}{100}$

Answer: 92%

Extension

$\frac{1}{3}$

1. $\times 100\% = \frac{100}{3}\%$

Answer: $33\frac{1}{3}\%$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.11

Percentage of a quantity

Date _____

Learning intentions

We are learning to:

- find a percentage of a quantity

Success criteria

I can:

- ☐ change the percentage to a fraction or decimal and multiply
- ☐ use 10% and 1% to find percentages mentally
- ☐ solve problems with percentages

Key terms Write the definition as your teacher explains it.

Discount

Percentage

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Find mentally.**a** Find 10% of 350.**b** Find 25% of 84.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find mentally.

12 QUESTIONS

REMEMBER Change % to a fraction or decimal.**1** 10% of 350

2 50% of 86

3 25% of 84

4 1% of 900

5 20% of 45

6 5% of 60

Questions 7–12: find mentally.

7 10% of 70

8 50% of 64

9 25% of 120

10 1% of 300

11 20% of 55

12 75% of 40

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: find mentally.

1 25% of 112

2 10% of 520

3 50% of 168

4 25% of 68

5 10% of 250

6 25% of 44

7 25% of 116

8 50% of 28

9 1% of 200

10 10% of 360

11 10% of 140

12 1% of 600

13 50% of 52

14 1% of 400

15 50% of 118

16 50% of 92

17 25% of 136

18 50% of 46

19 1% of 900

20 10% of 170

21 50% of 70

22 10% of 820

23 10% of 450

24 50% of 148

ROUND 2 Questions 25–40: find mentally. 5% is half of 10%. 20% is $2 \times 10\%$.

25 20% of 120

26 20% of 360

27 30% of 260

28 20% of 100

29 20% of 560

30 5% of 280

31 5% of 580

32 5% of 520

33 75% of 60

34 30% of 520

35 30% of 600

36 5% of 340

37 20% of 320

38 20% of 40

39 75% of 580

40 5% of 260

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 15% of \$60

4 75% of 36

2 7% of 1200

5 2% of 450

3 30% of 250

6 150% of 40



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find the percentage of the quantity.

a Find 15% of \$60.

Your teacher works this out. Copy every step.

b Find 7% of 1200.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Multiply by the quantity.

1 12.5% of 480 kg

3 8% of 2.5 km (m)

5 17% of 300

2 35% of \$180

4 45% of 3 h (min)

6 0.5% of \$2000

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: find the percentage of the amount.

1 60% of \$120

2 10% of \$360

3 85% of \$250

4 20% of \$330

5 60% of \$310

6 40% of \$220

7 95% of \$100

8 25% of \$290

9 15% of \$330

10 10% of \$230

11 75% of \$110

12 70% of \$50

13 50% of \$30

14 70% of \$370

15 90% of \$350

16 75% of \$70

ROUND 2 Questions 17–28: find the sale price.

17 \$130, 25% off

18 \$220, 10% off

19 \$260, 10% off

20 \$60, 10% off

21 \$40, 50% off

22 \$160, 30% off

23 \$120, 10% off

24 \$160, 15% off

25 \$270, 15% off

26 \$90, 50% off

27 \$180, 10% off

28 \$240, 15% off

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find, then solve the problem.

a Find 12.5% of 480 kg.

Your teacher works this out. Copy every step.

b A \$250 jacket is 30% off. Find the sale price.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Percentage problems.

4 QUESTIONS

REMEMBER 10%: $\div 10$. 1%: $\div 100$. Build others from these.

1 A \$250 jacket is 30% off. Find the sale price.

3 A \$40 meal has a 15% tip added. Find the total.

2 64% of 25 students caught a bus. How many did not?

4 Which is more: 30% of 70 or 70% of 30? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find 10% of 80.

Level 2

Find 35% of 200.

Level 3

A \$60 shirt is 15% off. Find the sale price.

Extension · Stretch yourself**Worked example**

A price rises by 10%, then falls by 10%. Is it back to the original price?

1. Try $\$100: + 10\% = \110
2. $- 10\% \text{ of } \$110 = \$110 - \$11 = \99

Answer: No: it is 1% lower

YOU DO

On your own: read the worked example, then try these.

- E1** A \$80 item has 20% off, then another 10% off the new price. Find the final price.

- E2** Is 20% off then 10% off the same as 30% off? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Change % to a fraction or decimal.
 2. Multiply by the quantity.
 3. $10\%: \div 10$. $1\%: \div 100$. Build others from these.
 4. Discount: subtract from the price.

Level 1

10% of 350

1. $350 \div 10$

Answer: 35

Level 2

15% of \$60

1. $10\% = \$6$, $5\% = \$3$

Answer: \$9

Level 3

30% off \$250

1. $30\% \text{ of } 250 = 75$
2. $250 - 75$

Answer: \$175

Extension

+10% then -10%

1. $100 \rightarrow 110 \rightarrow 99$

Answer: No, 1% less

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LESSON
4.12

Expressing as a fraction or percentage

Date _____

Learning intentions

We are learning to:

- express one quantity as a fraction or percentage of another

Success criteria

I can:

- ☐ write the part over the whole
- ☐ convert both quantities to the same unit first
- ☐ multiply by 100% to get a percentage

Key terms Write the definition as your teacher explains it.

Part

Whole

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write as a fraction and a percentage.

- a** Write 51 out of 60 as a fraction and a percentage.

Your teacher works this out. Copy every step.

- b** Write 18 out of 24 as a percentage.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write the first quantity as a fraction of the second, in simplest form. **12 QUESTIONS**

REMEMBER Write **PART** ÷ **WHOLE** as a fraction.

- 1** 6 out of 10 **2** 15 out of 20 **3** 51 out of 60 **4** 8 out of 12

- 5** 40 out of 100 **6** 9 out of 45

Questions 7–12: write as a percentage.

- 7** 3 out of 10 **8** 9 out of 20 **9** 12 out of 25 **10** 17 out of 50

- 11** 30 out of 40 **12** 4 out of 5

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write as a percentage.

1 2 out of 20
_____2 5 out of 10
_____3 3 out of 4
_____4 2 out of 4
_____5 1 out of 4
_____6 1 out of 5
_____7 2 out of 5
_____8 9 out of 10
_____9 23 out of 25
_____10 18 out of 50
_____11 3 out of 5
_____12 8 out of 25
_____13 2 out of 10
_____14 6 out of 20
_____15 11 out of 20
_____16 28 out of 50
_____17 21 out of 50
_____18 5 out of 20
_____19 5 out of 25
_____20 44 out of 50
_____21 7 out of 20
_____22 7 out of 10
_____23 4 out of 5
_____24 5 out of 50

ROUND 2 Questions 25–40: write the first amount as a fraction of the second, in simplest form.

25 6 out of 20
_____26 5 out of 16
_____27 15 out of 30
_____28 3 out of 20
_____29 17 out of 30
_____30 11 out of 24
_____31 58 out of 60
_____32 11 out of 20
_____33 13 out of 24
_____34 22 out of 40
_____35 44 out of 60
_____36 11 out of 12
_____37 1 out of 16
_____38 13 out of 16
_____39 18 out of 40
_____40 4 out of 12

YOU DO On your own: **Easy** Set B. Write as a percentage.

6 QUESTIONS

1 7 out of 10

4 13 out of 20

2 18 out of 24

5 45 out of 50

3 51 out of 60

6 3 out of 8

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Change to the same units, then write as a fraction and a percentage.

a Write 300 m as a fraction of 1.5 km.

Your teacher works this out. Copy every step.

b Write 300 m as a percentage of 1.5 km.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Change to the same unit, then write as a fraction and a percentage.

6 QUESTIONS

REMEMBER Use the same units for both.

1 300 m of 1.5 km

3 25 cm of 2 m

5 \$1.20 of \$4

2 45 min of 2 h

4 400 g of 5 kg

6 750 mL of 3 L

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: change to the same units, then write as a percentage.

1 200 g of 4 kg

2 50 c of 2 \$

3 10 c of 1 \$

4 10 c of 3 \$

5 100 m of 4 km

6 30 cm of 2 m

7 200 m of 1 km

8 500 m of 1 km

9 18 min of 3 h

10 750 g of 2 kg

11 15 min of 4 h

12 25 cm of 1 m

ROUND 2 Questions 13–28: test scores: write each as a percentage.

13 20 / 25

14 16 / 20

15 15 / 20

16 17 / 40

17 39 / 40

18 14 / 40

19 31 / 50

20 34 / 40

21 15 / 25

22 13 / 20

23 32 / 50

24 65 / 80

25 48 / 50

26 78 / 80

27 20 / 50

28 29 / 40

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write as a percentage, then compare.

a Write 45 min as a percentage of 2 hours.

Your teacher works this out. Copy every step.

b Mia scored 34/40 and Jake 42/50. Who did better?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER Simplify the fraction.

1 Mia scored 34/40 and Jake scored 42/50. Who did better?

3 A \$120 price dropped to \$90. What is the percentage decrease?

2 A team won 18 of 24 games. What percentage did they lose?

4 A class of 28 has 12 boys. What percentage are girls? (1 d.p.)

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 12 out of 16 as a fraction.

Level 2

Write 21 out of 25 as a percentage.

Level 3

Write 40 cm as a percentage of 2 m.

Extension · Stretch yourself**Worked example**

A shirt that cost \$40 now costs \$46. Find the percentage increase.

1. Increase = \$6
2. $\frac{6}{40} \times 100\%$

Answer: 15%

YOU DO

On your own: read the worked example, then try these.

- E1** A town grew from 2500 to 3100 people. Find the percentage increase.

- E2** A \$75 game is on sale for \$60. What percentage discount is this?

Lesson summary

Missed the lesson? Start here.

Remember

1. Write **part ÷ whole** as a fraction.
2. Use the same units for both.
3. Simplify the fraction.
4. Multiply by 100% for a percentage.

Level 1

51 out of 60

1. $\frac{51}{60}$
2. $\times 100\%$

Answer: 85%

Level 2

300 m of 1.5 km

1. 1.5 km = 1500 m
2. $\frac{300}{1500}$

Answer: $\frac{1}{5} = 20\%$

Level 3

45 min of 2 h

1. $\frac{45}{120}$

Answer: $\frac{3}{8} = 37.5\%$

Extension

\$40 → \$46

1. $\frac{6}{40} \times 100\%$

Answer: 15%

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